

Secure Data Communication

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Supervisor: Dr. Aung Htein Maw

Initial Contextual Report

COMP1682 Final Year Project

BSc (Hons) Computing (Network System)

Word Count: 9200 words

Abstract

This networking project proposed a 2-tier Network Architecture with IPSec site-to-site VPN for robust communication between two branches of an organization. The proposed network architecture prioritizes security and availability using encryption protocols, redundancies and safeguard mechanisms. While emphasizing on security of the network infrastructure, cost-efficiency and scalability are also taken into consideration. This proposal aims to deliver a comprehensive report of the technologies involved, and a network infrastructure that could facilitate uninterrupted workflow and secure communication between two branches of the organization while maintaining financial viability.

Acknowledgement

Firstly, I wish to acknowledge instructors from KMD Institute, especially Dr. Aung Htein Maw for his expertise in the networking field, and guidance throughout various aspects of this project. Secondly, I wish to acknowledge my parents, and uncle for their support and encouragement (both practically and emotionally). Lastly, I wish to acknowledge my classmates and friends. I appreciate them, and am thankful for being part of this journey together.

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1. Introduction

1.1 Background

TipTop is a typical business that has one main branch in Yangon, and one side branch in another city. Currently, the head quarter and the side branch are acting as two separate entities with little synchronization between them. In addition, communication between two branches is being done via generic consumer emails, phones and cloud services without any security assurances. As of now, both the headquarter and the side branch only got one main building each in the campus without any plans to expand in next 5-year timeframe. The business requires a network infrastructure that could accommodate **around 100 employees in each branch**, while also allowing the two branches to exchange data, communicate, perform day-to-day operations efficiently. It also wishes to establish a secure communication channel between the two branches to streamline the communication. Moreover, due to the prevalence of blackout in both Yangon and Mandalay, the business wants to ensure availability without any downtime.

1.2 Aims

The aim of the project is to *design* a tier-2 network architecture with IPSec and redundancy, *develop an implementation plan* and perform a simulation study of the internal network. The goal is to ensure secure and efficient data communication between TipTop's branches while maintaining reliability, availability and performance.

1.3 Objectives

1.3.1 Initial Research and Analysis [6.0]

- Research simulation tools (such as Packet Tracer, EVE-NG, GNS3 or NetSim) [2.0]
- Research about network infrastructure for businesses [4.0]

1.3.2 Research secure data communication and technologies [7.0]

- Research and analyse cyber threats [1.0]
- Research on networking technologies [3.0]
- Analyse and select suitable networking technologies [2.0]
- Literature Review [1.0]

1.3.3 Design and Implementation [10.0]

- Designing the business network architecture [6.0]
- Configuration of the technologies used [4.0]

1.3.4 Testing and Documentation [10.0]

- Testing the connectivity and security of the business network [3.0]
- Documentation and Evaluation [7.0]

1.4 Overall Issues of current communication model

TipTop currently is relying on consumer-grade products, and third-party services to manage internal communication and day-to-day operations. Although the products may suffice for personal or small office use, reliability, security and performance concerns are raised when considering for a business use. The lack of proper network infrastructure means the business is vulnerable to interruptions, connectivity, performance and reliability issues.

Furthermore, lack of advanced security and management feature exposes the business to cyber threats such as unauthorized access, and malware when communicating sensitive business information between two branches. Without segmentation or access control mechanism, internal network is unorganized and could face internal threats. Prevalent issues TipTop wishes to tackle include:

- Use of public ISP infrastructure without encryption leaves the business vulnerable to man-in-the-middle attacks
- Consumer-grade routers have limited capabilities and security features
- Lack of complete control for management and security due to the use of consumer or third-party solutions
- Lack of proper segmentation means one security breach could compromise the whole network

As the business expands, the network is expected to handle increased traffic, and improper network system could lead to performance and latency issues which, ultimately could impact the business. Overall, the reliance on consumer-grade products limits efficiency, and increases the risk of security, hindering the reliable and efficient operation of the business.

1.5 Implemented technologies

Following network technologies are implemented in network architecture for TipTop business.

- **IPSec VPN:** Main focus of the project; ensures encrypted, authenticated communication between branches.
- **Redundancy:** high availability; ensures the network stays up and running even if a device or link fails
- **Dynamic Routing Protocol (OSPF):** dynamically set most efficient route between networks, eliminating the need for manual routing.
- **VLANs:** logically divide a network into segments for management, security and efficiency.
- **Subnetting:** dividing large networks into manageable pieces; decreasing broadcast traffic, increasing latency and enhancing performance.

2. Literature Review

2.1 Approach for Research

The literature search involved using Google and textbooks to access a wide range of web-based resources on the theoretical and practical knowledge into IPSec VPN and Network Architecture implementation. Google was extensively used to access official documentation from vendors such as Cisco, other resources such as tutorials, and videos. *Computer Networks* by Andrew Tanenbaum and *A Top-Down Approach* books were used for **theoretical and in-depth understanding of the IPSec VPN**. These sources were referenced to support structured research and implementation of this project.

2.2 Research on Cyber Threats

Business can be targets for cyber-attacks due to the valuable data they have. This section will briefly discuss about common threats, how to prevent them and what to do when attackers got access. Below are some common threats businesses usually face, along with some preventative measures to counter them.

Phishing Attacks

Malicious individuals could trick employees of the business into revealing sensitive information data of the business by pretending to be trusted individuals, or leading them to do harmful actions. Some methods for phishing attacks include-

- Emails containing malicious files, or links
- Social engineering attacks
- Phishing through text messages, or SMS, or
- Whaling, targeting high level executives who have access to classified information (fortinet, n.d.)

These threats could be counter by implementing email filtering, training employees to be cognizant about potential threats, and using multi-factor authentication to protect accounts.

Ransomware Attacks

Malware that encrypts important business data and asks for ransom might creep their way into the system via suspicious emails or download links. Employees should not click on suspicious links or download unvalidated files from websites. This could be prevented by only allowing legitimate websites in the firewall, and using access-control mechanisms for installing software on the network.

Insider Threats

Intentional or unintentional misuse of access to resources by employees. This type of threat may be malicious or negligent in nature. Preventative measures for this type of threat include audit logs, access-control mechanisms, and enforcing offboarding procedures, for example NDA (IBM, n.d.).

Distributed Denial of Service Attacks (DDoS)

Malicious attackers may flood the network with traffic while denying legitimate users of the access. This compromises the availability of the service, by causing slowdowns or crushing the system entirely. Configuring rate-limiting and traffic-filtering on firewalls/routers, or using DDoS protection may mitigate this attack.

In additional to these traditional means, employees must be informed about newly emerging threats such as

- **Rubber Ducky:** innocent looking USB drive with malicious payload that can emulate USB keyboard to attack workstations (plurilock, n.d.)
- **O.MG cables:** which looks like ordinary phone charger cable, but could connect to nearby Wi-Fi access point to execute remote payloads, including keystrokes injection and device control (triskelelabs, n.d.)

Despite all the preventative measures, when attackers got their hands on data, it is important contain the extent of damage that could be done to the business using breached data.

2.3 Research on Simulation Tools

Network Simulation tools are essential for studying and designing network topologies without any hardware requirements, and consideration for physical limitations. There are many network simulation tools each with their own unique advantages and disadvantages. Following simulation tools are considered along with their vendor-availability (i.e., different manufacturers of networking devices):

1. Packet Tracer (Cisco devices only)
2. EVE-NG (multi-vendor)
3. GNS3 (multi-vendor)
4. Tetcos NetSim (vendor-neutral)
5. Cisco Modelling Lab (Cisco devices only)

Packet Tracer

Packet Tracer is a light weight simulation tool developed by Cisco, which allows users to create network designs and simulate network systems. The software can be run on various operating systems, including Linux, Windows, macOS, and even mobile platforms (albeit limited) such as Android and iOS. The intuitive user-interface and drag-and-drop feature for the devices of the packet tracer lowers the barrier to entry and allow students to familiarize themselves with modern networking devices and technology (sri, 2020).

From personal experience, Packet Tracer can support small to medium networks without any performance issues (although some bugs exist). Although Packet Tracer could reproduce most of the functions of Cisco networking devices, literal simulation is not supported, resulting in limited functionality for some advanced features.

Advantages	Disadvantages
- Free and light-weight - User-friendly - Can support medium networks - Runs on multiple Operating Systems	- Cisco device only - Does not support complex and multi-vendor networks - Some Bugs

EVE-NG

EVE-NG (Emulated Virtual Environment Next Generation) is a multi-vendor network emulation tool that allows virtualization of networking hardware and software. It offers emulation of multiple devices, and easy integration of external tools (Patel, 2024). It is based on Linux (Ubuntu) and could be installed on bare-metal hardware, virtual machines (VMs) or on cloud. Unlike Packet Tracer, it emulates actual devices and could provide actual functionalities of the network devices (Sims, 2023).

There are two versions of EVE-NG; professional and community. The professional version costs \$170 per year, while the community version is free, with some trade-offs. The free version is limited in node and lab capacity, and lacks some features such as not being able to move nodes without switching them off (a deliberate and needless limitation). The images for networking devices in EVE-NG have to be acquired directly via the vendors, or from physical devices, which limited the accessibility.

Although it supports multiple vendors, and complex networks, it is resource intensive and requires a virtual machine (VM) or a dedicated server for ideal usage. The setup process before usage may not be intuitive as it involves several technical steps.

Advantages	Disadvantages
- Can emulate and provide actual network devices - Clean and User-Friendly Interface - Web-based UI - Support complex and multi-vendor networks	- Cost Efficiency - Resource intensive - Requires technical knowledge for installation process

GNS3

Graphical Network Simulator 3 (GNS3) is an open-source network simulation tool widely used by network professionals and students due to its extensibility, community support for documentation and trouble-shooting, and realism. GNS3 and EVE-NG have similarities and one of them being the ability to emulate actual images of networking devices. In addition, images for networking devices (for example, Cisco IOS) requires proper licensing, or have to be acquired from physical devices, adding complexity (Sims, 2023).

Deployment options for GNS3 include directly on the PC as a software, in Virtual Machine, or on Clouds. Unlike EVE-NG however, GNS3 is completely free and rely on community support, thus making it cost efficient (Sims, 2023).

Despite the ability to be directly used on local machine as a software, for better performance and scalability, the usage of VM or cloud is recommended, thus making it resource intensive.

Advantages	Disadvantages
- Can emulate and provide actual network devices - Free to use - Multiple deployment options - Support complex and multi-vendor networks	- Resource intensive (if used on VM) - Requires technical knowledge for setup process

NetSim

Tetcos NetSim (not to be confused with Boson NetSim) is a powerful tool for protocol modelling and network simulation designed for research and academic learning. It allows users to design and simulate network topologies, and perform high-level analyses for protocols, network metrics, and devices in a controlled virtual environment. It supports industry-standard protocols such as TCP, UDP, FTP, 5G, HTTP, Wi-Fi, LTE, and IoT.

Tetcos NetSim could provide valuable insights into network metrics and protocol-specific analysis. However, it is not ideal for this project as it is research oriented and might not be suitable for designing and simulating real-world devices.

Advantages	Disadvantages
- Can provide comprehensive outputs for analysis in the form of packet-logs, load graphs and visualized metrics - Supports wide range of in-depth protocol analyses - Integration support (for example Wireshark and MATLAB)	- Platform limitation (only on Windows) - Cost - Research Oriented

Cisco Modelling Lab (CML)

Cisco Modelling Lab is a premium alternative to Cisco Packet Tracer that includes genuine Cisco device images without having to install them manually. Although Cisco offers a free version of CML to individuals, it is limited to five active nodes, impractical for this simulation. It is based on Linux and can be run either on Virtual Machines or on Cloud. Unlike Packet Tracer, it can run complex network designs, and fully supports switching and routing protocols, security configurations and other advanced technologies. Packet capture tools are also integrated for real time analysis and trouble-shooting. CML for pricing for individual costs \$200, with built-in Cisco IOS images (Cisco , n.d.).

Advantages	Disadvantages
- Built-in Cisco IOS images - Support complex networks - Include real-time packet capture	- Resource intensive and requires VM or Cloud - \$200 cost - Free version only supports up to 5 nodes

Conclusion

Due to Cisco Market dominance, familiarity and to maintain the relevancy of this project, Cisco devices will be used. However, all the simulation tools, aside from NetSim could accommodate this requirement, thus the choice is made based on factors such as cost, complexity and efficiency.

Despite the limitations, Packet Tracer will be used for simulation for this project as most protocols and features that will be used in this project could be simulated on it, with free-to-use, readily available built in devices (does not require manual image installation) and lightweight-ness as bonuses.

2.4 Research on Network Architecture

LAN Architectures

Architectures are standardized ‘best practice’ when designing a network, and requirement for each business may vary. It is important to analyse and understand the specific needs of a business before designing a network. Common types of architectures are 2-tier, 3-tier and Spine-Leaf architecture.

Following are common terminologies that will be used.

Star Topology - In Star topology, all devices are *only connected to one central device* in a ‘star’ shape, without connecting to each other. However, in practice, the topology might be similar to fig 2.4.2.

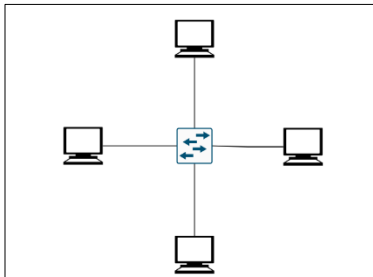


Fig 2.4.1 Star Topology

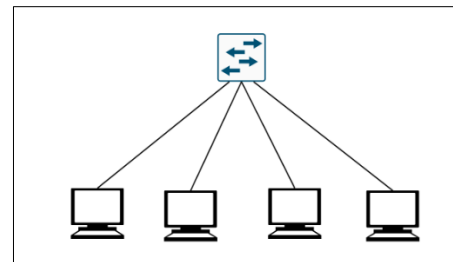


Fig 2.4.2 Star Topology (in practice)

Full Mesh - Full Mesh is when all devices are all connected to all the other devices as shown in fig 2.4.3.

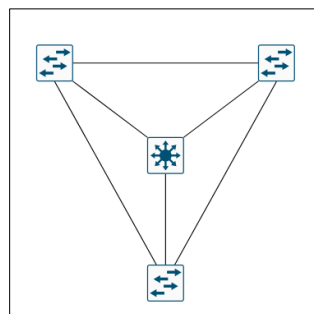


Fig 2.4.3 Full Mesh

Partial Mesh – When devices are connected to each other, but not all of them are connected.

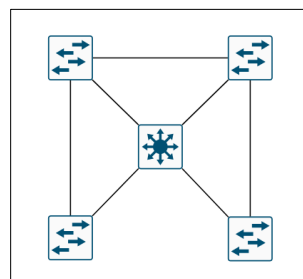


Fig 2.4.4 Partial Mesh

2-Tier and 3-Tier Architecture

2-tier and 3-tier are traditional LAN architecture used in enterprise networks. These architectures are common in businesses where the traffics are usually North-South (i.e., traffics travels from end devices to access layer, then to distribution layer, etc and eventually out of the LAN) (geeksforgeeks, 2022).

2-Tier Architecture

Consists of Access and Distribution Layers. As it combines the Core Layer found in 3-tier design into one layer, it is also called ‘Collapsed Core’ design. **The Access Layer** consists of switches that are connected to end devices (PCs, Laptops, printers). As this layer consists of layer-2 switches, Spanning Tree Protocol (STP) is used to prevent loops in this layer. The Access Layer switches are then connected to a redundant pair of switches in **the Distribution Layer**. Multi-layer switches are usually used for this layer. These multilayer switches use Switch Virtual Interface (SVI) to act as default gateway for the end devices, and First Hop Redundancy Protocols (such as HSRP) for redundancy.

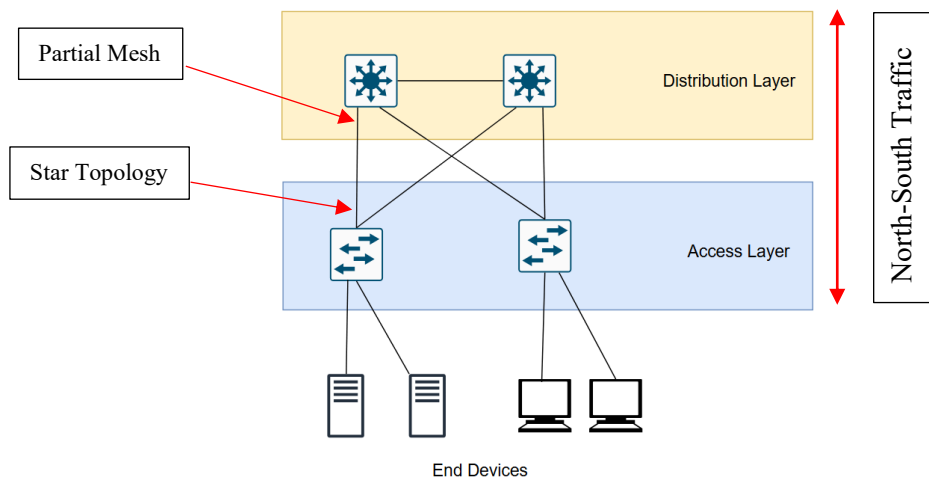


Fig 2.4.5 Two-Tier Architecture

3-Tier Architecture

In a large LAN, where there are two or more distribution layers (for example, 2 or more separate buildings in a location), the distribution layers should be connected to **Core Layer** for scalability and efficiency. Fast link speed is the primary goal of the core layer and upward of **10 Gigabit ethernet** is recommended (geeksforgeeks, 2022). CPU intensive workloads such as security and QoS marking/classification should not operate at this layer.

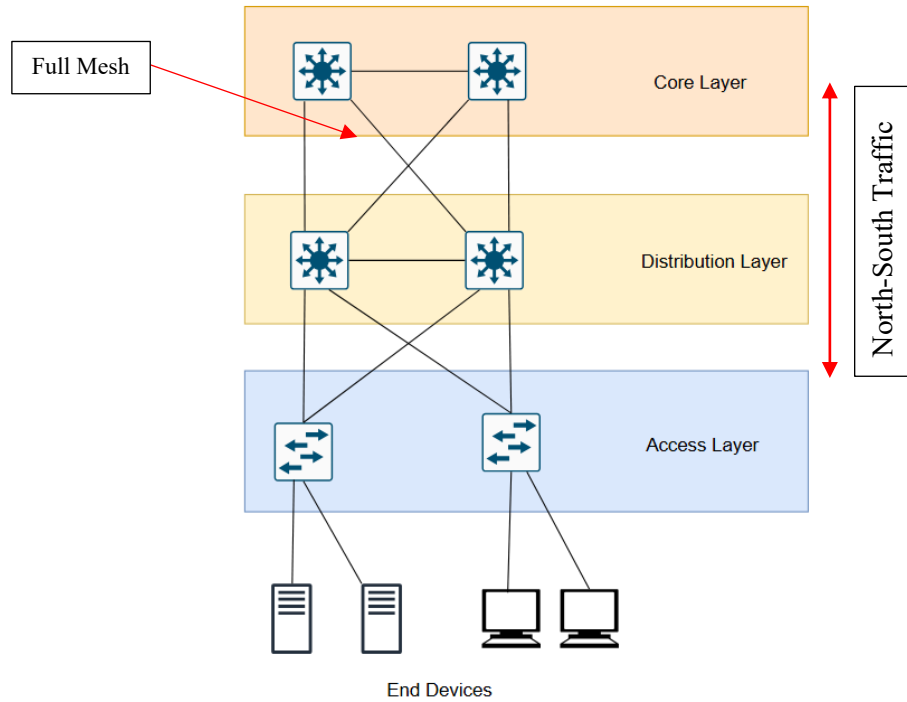


Fig 2.4.6 Three-Tier Architecture

Spine-Leaf Architecture

In Spine-Leaf architecture, the traffic is east-to-west rather than north-south as in 2-tier and 3-tier architecture. It is consisted of Leaf Layer and Spine Layer. The Leaf Layer, similar to Access Layer, is connected to the end devices. Each of leaf switches of this layer is connected to spine switches. However, leaf switches **do not connect** to other leaf switches, and spine switches **do not connect** to other spine switches.

The path taken by traffics in this architecture is randomly chosen to load-balance between the leaf switches (geeksforgeeks, 2022). This architecture is *highly scalable* as more leaf switches could be added as needed to connect to more end devices. The newly added leaf switch could then be linked to one of the spine switches.

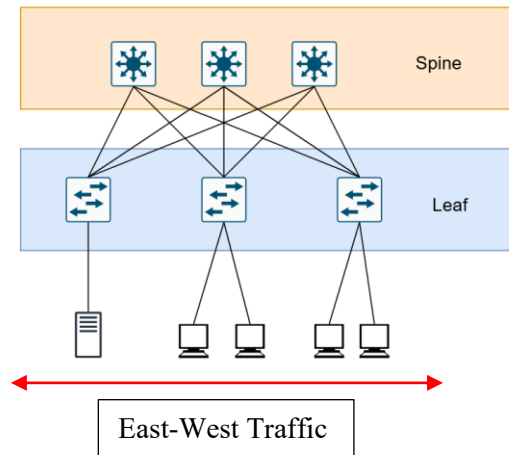


Fig 2.4.7 Spine-Leaf Architecture

Comparison between Architectures

	Spine-Leaf	2-tier	3-tier
Traffic Flow	East-West	North-West	North-West
Scalability	Highly Scalable	Moderately Scalable	Less Scalable due to multiple layers
Latency	Low (fewer hops)	Moderate	High
Redundancy	High redundancy	Moderate redundancy	High redundancy
Cost	High upfront cost	Cost effective for small to medium business	Expensive due to multiple layers
Ideal Use Case	Data Centres	Small to medium business	Large business

Conclusion

After the analysis and considerations for financial viability, practicality and efficiency, the 2-tier architecture emerges as the ideal choice for TipTop.

Although the other architectures could offer higher redundancy, more scalability and lower latency, 2-tier could offer simple yet scalable design that aligns with the operational needs of TipTop. Furthermore, as both headquarter and the side branch of TipTop operate within a single building, each branch requires only one distribution layer, without the need for core layer.

It could provide sufficient redundancy and reliability for a medium sized business without overcomplicating the design. It is cost-efficient as it requires less devices and cabling compared to 3-tier and Spine-Leaf architectures.

In conclusion, the 2-tier architecture strikes the balance between performance, cost-efficiency and simplicity for secure data communication between two branches of TipTop.

2.5 Research on Routing protocols

Routing

Routing is the process to determine which path is the most efficient for network packets to take to reach to the destination. There are two main routing methods; *static routing* where network routes on routers are manually configured, and *dynamic routing* using protocols (such as OSPF) which shares the routing information automatically and build their own routing tables. In this project, with the amount of routing devices that will be used in the infrastructure, dynamic routing is ideal for efficiency and scalability in the future.

Dynamic Routing

Dynamic Routing is divided into two categories; Interior Gateway Protocol (IGP) and Exterior Gateway Protocol (EGP). IGP is used within a single autonomous system (i.e., within one organization or company), and EGP is used to share routes between different autonomous systems.

As the two branches of TipTop will be connected over a private tunnel with the use of site-to-site VPN, only internal routing protocols will be considered.

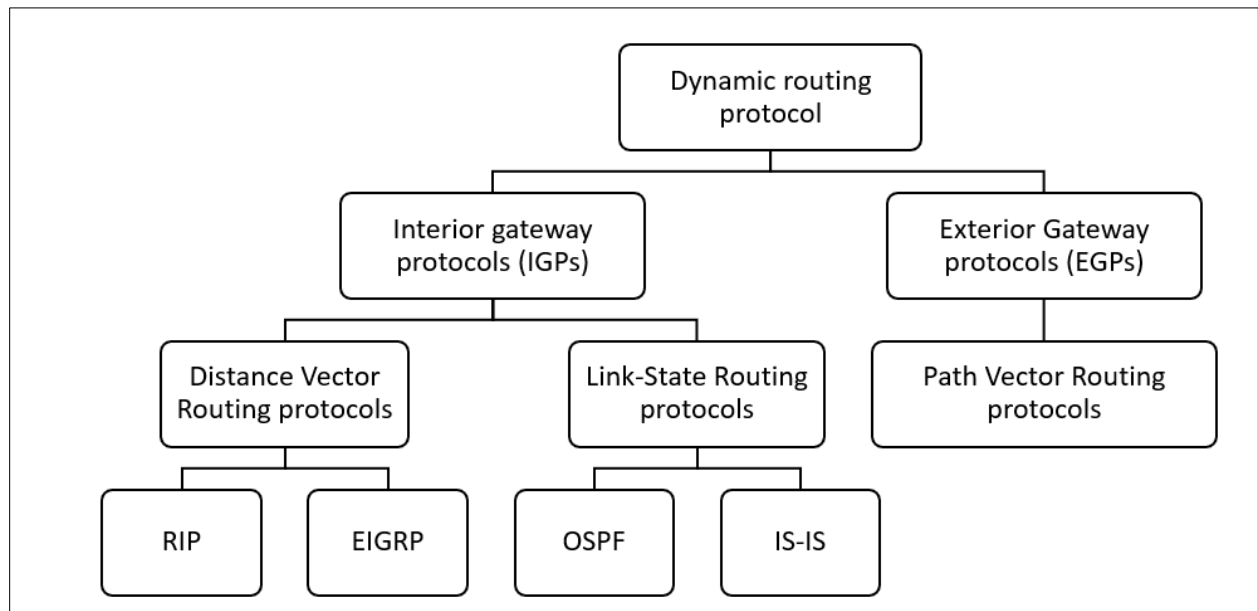


Fig 2.5.1 Dynamic Routing Protocols

Interior Gateway protocols (IGPs)

There are two types of IGP that uses different algorithms; Distance Vector routing algorithm, and Link State routing algorithms.

Distance Vector Routing Protocols include:

- Routing Information Protocol (RIP)
- Enhanced Interior Gateway Routing Protocol (EIGRP)
(Interior Gateway Routing Protocol is obsolete by modern standards and will not be discussed)

Link State Routing Protocols include:

- Open Shortest Path First (OSPF)
- Intermediate System-to-Intermediate System (IS-IS)

Route Selection

Routers choose the route with longest prefix length whenever there is a need to forward a traffic for optimal performance. For example, a traffic with destination to 10.0.0.5 has arrived. Between two routes **10.0.0.0/24** and **10.0.0.5/32**, the router would choose the route with /32 over the /24 one.

Should the prefixes of the routes be the same, it evaluates the Administrative Distance (AD) to determine the most reliable route. If multiple routes have the same prefix and AD, it then considers the metric to choose the most efficient path. Different routing protocols have different metrics.

longest Prefix → lower Administrative Distance → lower Metric

Administrative Distance (AD)

Different routing protocols have different Administrative Distances, and lower AD is preferred as it indicates the route is 'more trustworthy'. Below are the list of route types/protocols and their respective AD values.

Route Type/Protocol	AD	Route Type/Protocol	AD
Directly connected	0	IS-IS	115
Static	1	RIP	120
EIGRP	90	EIGRP (external)	170
IGRP	100	Unusable Route 255	
OSPF	110		

Routing Information Protocol (RIP)

RIP is one of the oldest dynamic routing protocols and uses distance vector algorithm. It uses *hop count* as metric to determine the best path to destination, with *hops* being the number of routers traversed. Maximum number of hop counts for a path is 15. Due to ease of implementation and simplicity, it is used in small business networks, but with the maximum hop counts limited to 15, it is not suitable for larger and complex networks (zenarmor, 2024). There are three versions of Routing Information Protocols:

- RIPv1: supports only classful routings, and uses broadcast to send updates.
- RIPv2: supports classless routing, and uses multicast to send updates.
- RIPv6: similar to RIPv2 but also supports IPv6.

Advantages	Disadvantages
- Simple and Easy configuration - Widely supported - Less usage of CPU and memory resources	- Limited scalability (15 max hops) - Slow convergence - High bandwidth usage - Security Issues

Enhanced Interior Gateway Routing Protocol (EIGRP)

Enhanced Interior Gateway Routing protocol is a hybrid protocol that combines features of both link state and distance vector protocols. It was developed by Cisco in the early 90s and mostly used Cisco environment. Although EIGRP has become ‘open standard’ in 2013, it is not as prevalent as open standard OSPF. EIGRP is the only protocol that supports Unequal Load balancing (zenarmor, 2024).

Although EIGRP ensure fast convergence by using Diffusing Update Algorithm (DUAL), it requires more memory and processing power. EIGRP is more complex compared to RIP as it requires delicate configuration of metrics, route summarization and authentication (zenarmor, 2024).

Advantages	Disadvantages
- Fast Convergence - Load balancing (only protocol that supports unequal load balancing) - Efficient Bandwidth Usage	- Cisco Proprietary (mostly) - Common only in Cisco environment - Excessive resource utilization - Complex configuration

Open Shortest Path First (OSPF)

OSPF (Open Shortest Path First) is a link-state routing protocol that could handle large and complex networks efficiently. It is an open standard and works across multi-vendor network environments. OSPF uses Dijkstra’s algorithm to calculate the most efficient route and supports hierarchical network designs by the use of *areas*. Moreover, OSPF also ensures effective bandwidth usage as it sends updates when only changes occur, maintaining scalability. Not only is OSPF efficient in directing IP packets to their destination, it also has the ability to detect any changes in the topology (zenarmor, 2024). To enhance security, it uses MD5, SHA-256, and plain text authentication, protecting routing updates from attacks.

Advantages	Disadvantages
- Scalability (supports large networks and not limited to 15 hops like RIP) - Efficient use of Bandwidth - Open-standard and multi-vendor compatible - Supports Route authentication	- High CPU and Memory usage - Complex configuration (unlike plug-and-play like RIP) - Does not support Unequal-cost load balancing unlike EIGRP

Intermediate System-to-Intermediate System (IS-IS)

Intermediate System-to-Intermediate System (IS-IS) is a link-state routing protocol similar to OSPF. It operates at the Network Layer, but unlike OSPF which runs directly over IP, it uses CLNS (Connectionless Network Service) for transportation. Later, integrated IS-IS is developed to support IP routing (networklessons, n.d.) (zenarmor, 2024). IS-IS is preferred in large ISP networks due to its scalability and efficient routing updates.

IS-IS uses two-level routing: level-1 and level-2 routing. Level-1 operates within an intra-area, while level-2 is similar to backbone area in OSPF, as it is connected to all leve-1 intra-areas. Leve-1 routers can only communicate within same area, and to connect between level-1 and level-2 areas, leve-1/2 routers are used as liaisons between two levels (zenarmor, 2024).

Advantages	Disadvantages
- Highly Scalable (suitable for large ISPs) - Supports multi-level hierarchy (better than OSPF) - Less commonly attacked compared to OSPF (as it is not as popular)	- More complex configurations - Limited Vendor Support - No native load balancing - Less popular (hard to find trained engineers)

Comparison

Feature	IS-IS	OSPF	EIGRP	RIP
Type	Link State	Link State	Hybrid (DV+LS)	Distance Vector
Metric	Cost (Bandwidth)	Cost (Bandwidth)	Bandwidth + Delay	Hop count
Convergence	Fastest	Fast	Fastest	Slow
Scalability	Best for ISP and large Networks	Good for large networks and multi-area support	Good for medium sized networks	Not Good (Limited to 15 hop counts)
Compatibility	Yes (Most vendors)	Yes (open standard)	Mostly Cisco	Yes (open standard)
Security	Strong	Strong (MD5, SHA)	Strong (MD5, SHA)	Weak

Conclusion

OSPF is the most popular routing protocol in the industry due to its popularity, efficiency, security and multi-vendor compatibility. Among the four routing protocols discussed above, OSPF is ideal for TipTop for the following reasons:

Efficiency and Reliability: unlike RIP which sends periodic updates, OSPF only sends updates only when there are changes, reducing bandwidth utilization and improving response time. OSPF is more scalable, as RIP is limited to 15 hop counts.

Compatibility: although Cisco devices will be used for this TipTop project, using popular and *fully* open-standard OSPF has its advantages for future compatibility, where the company might decide to change to different vendor devices.

Easy Configuration: although both IS-IS and EIGRP have faster convergence and are more efficient, their complex configuration can be a limiting factor, especially they are not as commonly used as OSPF in enterprise networks. Thus, finding competent engineers who could fine tune these technologies can be a challenge.

For these reasons, OSPF is chosen as the ideal routing protocol for TipTop network project. Further details on OSPF will be discussed in Design and Implementation section.

3. Legal, Social, Ethical and Professional Issues

3.1 Legal

Privacy and Data Protection – Handling users’ information in compliance with data protection laws, such as GDPR, is essential, as non-compliance could result in legal penalty or loss of customers’ trust. For example, storing users’ information without encryption or consent would violate data protection laws.

Intellectual Property (Rights) – Network designs, software and configurations must not infringe copyrights and patented technologies. TipTop must comply with licensing agreements for proprietary technologies such as firewall configurations, and Cisco IOS images.

Cybersecurity Regulations – TipTop must abide by standards and cybersecurity laws imposed by regulatory bodies or government organizations to prevent data breaches and cyberattacks. It must implement access control mechanisms and incident response plans to comply with the regulations. For example, in implementing the IPSec VPN between two branches of TipTop, it must follow strong encryption standards.

3.2 Social

Accessibility – not all users may have access to high-speed internet or network technologies, and the TipTop must consider how network solutions may impact the users in remote regions or without access to the internet.

Social Diversity and Inclusion – the business must treat all the employees of different race, background and gender are treated equally, and discriminations and hate speech should be prohibited to ensure positive workplace environment.

Privacy Concerns – employees of TipTop expect their personal data to be secure, and privacy to be respected. Excessive monitoring of employees in the network may violate employees’ privacy. The business must be transparent about their monitoring and data collection policies.

Security Awareness – having proper cybersecurity training and awareness is essential for keeping the business and its network secure. Despite the latest security technologies and strong encryptions, a lack of employee awareness could render them ineffective as “the security is only as strong as the weakest link”. Mandatory trainings for employees and proper security procedures should be implemented for mitigating the issue.

3.2 Ethical

Confidentiality and Data Usage – the business is legally and ethically responsible for keeping personal information of its users safe. Improper sharing of data or negligence in security could be seen as irresponsible. For example, TipTop sharing its users' data with other organizations without prior consent could be seen as unethical conduct.

Energy Efficiency and Sustainability – businesses have ethical obligation to reduce carbon-footprint by using energy efficient network devices and adopt greener energy, without compromising on the performance. For example, choosing power efficient devices, and integrating renewable energy solutions such as solar power for the network infrastructure.

3.3 Professional

Responsibility and Accountability – TipTop should clearly define roles and responsibilities of each employee in case of network failures or security breaches. Moreover, the professionals in charge of the network should be diligent and responsible in maintaining the business network. For example, if a network engineer misconfigured firewall access-list, leading to security vulnerabilities, they must take responsibility and fix the issue immediately.

Professional Development and Certifications – network technologies are evolving and the professionals responsible for TipTop network must strive continuous development, and be aware of new trends, threats and protocols. Taking certifications such as CCNA, CCNP could help professionals stay updated with current industry standards.

4. Design and Implementation

4.1 Implementation of Network Infrastructure

Network Architecture Design

As discussed above (in topic 2.4), 2-tier Architecture is ideal for TipTop due to its cost-efficiency and performance, following Architecture is designed for each branch. This architecture will be able to accommodate around one hundred employees in each branch, while also being easily scalable with the addition of new **access switches**.

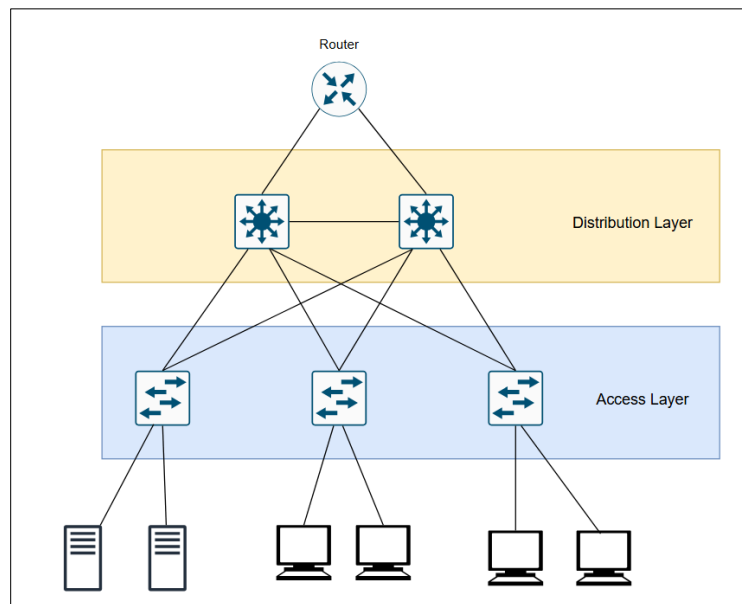


Fig 4.1.1 2-tier network architecture for each TipTop branch

Two multilayer switches will be used in the **distribution layer**, configured with HSRP for redundancy and SVI to act as one virtual router for the access layer switches.

Three layer-2 access switches, each with 48 ports, will be connecting the distribution layer and the end devices, supporting up to roughly 100 users across the network. Spanning Tree Protocol (STP) will be implemented to prevent loops, and PortFast enabled on the interfaces connecting to the end devices to improve connectivity and performance.

At the top of the architecture is a router connecting the entire LAN to the Internet, and to the other branch via the IPSec VPN, allowing secure and reliable communication between sites.

(Disclaimer: end devices connected in Fig 4.11 may not represent the actual network configurations)

Network Hardware Device Implementation

In Fig 4.1.1 network architecture, following hardware are required:

- 1 Router
- 2 Multilayer Switch
- 3 Layer-2 Access Switch (initially, could be added for scalability)
- Cat6 cables

As discussed in topic 2, Cisco devices will be used. Following Cisco devices, that are considered for the TipTop network infrastructure, along with their prices and links.

(As the business is located in Myanmar (Burma), local currency is MMK, but for sake of relevancy and simplicity USD is used. The exchange rate as of February 26th 2025 is 1 USD = 4,500 MMK.)

Type	Name	Description	Price	Link																								
Router	C1111-4P	• PORT COUNT: 4-port managed Gigabit Ethernet switch (2 ports PoE capable with 1 port PoE+) • CONNECTIVITY: Gigabit WAN and WAN/SFP combo ports with integrated USB 3.0 AUX/console • SECURITY: Multi-layer, robust security, and high-performance VPNs encrypt all data from the WAN edge to the cloud • PERFORMANCE: Business router designed for broadband network speeds while running secure, concurrent data, voice, video, and wireless services • PEACE OF MIND: 1-year limited hardware warranty with 10-day advanced replacement	\$ 995	Click Here																								
	K9-ISR 4331	<table><tr><td>Brand</td><td>Cisco</td><td>Included Components</td><td>Accessories, Router</td></tr><tr><td>Model Name</td><td>ISR4331/K9</td><td>Connectivity Technology</td><td>Ethernet</td></tr><tr><td>Special Feature</td><td>Integrated Services Router</td><td>Color</td><td>Black</td></tr><tr><td>Frequency Band Class</td><td>Tri-Band</td><td>Antenna Type</td><td>Internal</td></tr><tr><td>Compatible Devices</td><td>Personal Computer</td><td></td><td></td></tr><tr><td>Recommended Uses For Product</td><td>Business</td><td></td><td></td></tr></table>	Brand	Cisco	Included Components	Accessories, Router	Model Name	ISR4331/K9	Connectivity Technology	Ethernet	Special Feature	Integrated Services Router	Color	Black	Frequency Band Class	Tri-Band	Antenna Type	Internal	Compatible Devices	Personal Computer			Recommended Uses For Product	Business			\$ 895	Click Here
	Brand	Cisco	Included Components	Accessories, Router																								
Model Name	ISR4331/K9	Connectivity Technology	Ethernet																									
Special Feature	Integrated Services Router	Color	Black																									
Frequency Band Class	Tri-Band	Antenna Type	Internal																									
Compatible Devices	Personal Computer																											
Recommended Uses For Product	Business																											
Cisco ASR1002	<table><tr><td>Brand</td><td>Cisco</td><td>Connectivity Technology</td><td>Wired</td></tr><tr><td>Model Name</td><td>ASR1002</td><td>Item Weight</td><td>44 Pounds</td></tr><tr><td>Special Feature</td><td>WPS</td><td>Operating System</td><td>Cisco IOS</td></tr><tr><td>Compatible Devices</td><td>Personal Computer</td><td>Security Protocol</td><td>WPS</td></tr><tr><td>Recommended Uses For Product</td><td>Business</td><td>Number of Ports</td><td>4</td></tr><tr><td>Connectivity Technology</td><td>Wired</td><td></td><td></td></tr></table>	Brand	Cisco	Connectivity Technology	Wired	Model Name	ASR1002	Item Weight	44 Pounds	Special Feature	WPS	Operating System	Cisco IOS	Compatible Devices	Personal Computer	Security Protocol	WPS	Recommended Uses For Product	Business	Number of Ports	4	Connectivity Technology	Wired			\$ 1,026	Click Here	
Brand	Cisco	Connectivity Technology	Wired																									
Model Name	ASR1002	Item Weight	44 Pounds																									
Special Feature	WPS	Operating System	Cisco IOS																									
Compatible Devices	Personal Computer	Security Protocol	WPS																									
Recommended Uses For Product	Business	Number of Ports	4																									
Connectivity Technology	Wired																											

Type	Name	Description	Price	Link									
Multilayer Switches	Catalyst 9300L-48P-4X-E	• Features 48 networking ports to establish large network • Lets you conveniently connect different networks to enable improved flexibility to your infrastructure • Stack port provides an easy and hassle-free way to extend your office network • Optical Fiber media support for a dependable and productive connectivity • Optimize the performance of business networks with 10 Gigabit Ethernet connectivity • Can work as layer 3 routing for scalable network design • Managed switch provides a reliable foundation for your network	\$ 4,271	Click Here									
	C9200L-24T-4G 24 Port	• Total Number of network Ports: 24 • Uplink port: Yes • Modular: Yes • Stack port: Yes • Port/expansion slot details: 24 x Gigabit Ethernet network • Memory storage capacity: 2000 MB	\$ 960	Click Here									
	N9K-C93240YC-FX2 (renewed)	<table><tr><td>Brand</td><td>Cisco</td></tr><tr><td>Number of Ports</td><td>48</td></tr><tr><td>Included Components</td><td>Switch, Accessory Kit, Menu Instruction</td></tr><tr><td>Compatible Devices</td><td>Storage, Server</td></tr><tr><td>Switch Type</td><td>fixed, managed, cut-through</td></tr></table>	Brand	Cisco	Number of Ports	48	Included Components	Switch, Accessory Kit, Menu Instruction	Compatible Devices	Storage, Server	Switch Type	fixed, managed, cut-through	\$ 1,860
Brand	Cisco												
Number of Ports	48												
Included Components	Switch, Accessory Kit, Menu Instruction												
Compatible Devices	Storage, Server												
Switch Type	fixed, managed, cut-through												

Type	Name	Description	Price	Link
L2 Access Switches	WS-C2960S-48TS-L	• 48 x 10/100/1000 Ethernet ports • 4 x SFP (Small Form-factor Pluggable) uplink ports • 1 RU (Rack Unit) form factor • Stackable switch with FlexStack technology (optional module required) • Switching Capacity: 176 Gbps • Forwarding Rate: 41.7 Mpps • 128MB Flash Memory • 256MB DRAM	\$ 1,500	Click Here
	C1000-48T-4G-L	• HARDWARE CONFIGURATION: 48 10/100/1000 Gigabit Ethernet ports, 4 SFP+ uplink ports • MANAGEMENT: Manage the switch via the Cisco IOS CLI (command-line interface) and/or intuitive web UI. Centrally monitor and configure up to 8 switches via a single IP or with any third-party network management tool • SECURITY: Enterprise-class security that natively supports IPv4/IPv6 FHS security features such as port security, DHCP snooping, and dynamic ARP inspection, which can easily be configured using a single "Auto-Secure" command, along with 802.1x support for all connected devices • PEACE OF MIND: Enhanced limited lifetime warranty	\$ 900	Click Here
	CBS250-48T-4X	• SWITCH PORTS: 48 -Port 10/100/1000 + 4 x 10GE SFP+ • SIMPLE: Intuitive Cisco Business Dashboard or on-box U/I simplifies network operations and automates lifecycle management • FLEXIBLE: Extensive portfolio provides ultimate flexibility from 8 to 48 ports and PoE+ combinations • SECURITY: Integrated with IEEE 802.1X port security to control access to your network, Denial-of-Service (DoS) attack prevention increases network uptime during an attack, while extensive Access Control Lists (ACLs) protect the network from unauthorized users and guard against network attacks • INNOVATIVE DESIGN: Elegant and compact design, ideal for installation outside of wiring closet such as retail stores, open plan offices, and classrooms	\$ 700	Click Here

Chosen Network Devices

In order to implement for both branches, **the amount of hardware needs to be doubled**. Of the listed Cisco devices above, following are chosen:

Name	Type	Reason	Price	Quantity	Link
C1111-4P	Router	- Supports IPSec VPN for site-to-site - Built-in security features - 4 gigabit ethernet ports - Energy efficient and consumes less power - Good price-to-performance	\$ 995 (x2)	1 (x2)	Click Here
C9200L-24T-4G	L3 Switch	- Supports Energy Efficient Ethernet (EEE) for power savings - Layer-3 functionality - 4 uplink ports to connect to router, firewalls, or WAN ports - 24 Gigabit ports for future scalability	\$ 960 (x4)	2 (x2)	Click Here
CBS250-48T-4X	L2 Switch	- Cost efficient - 48 ports - STP and PortFast Support - Energy Efficient	\$ 700 (x6)	3 (x2)	Click Here
Total Cost			\$10,030		

Utility Tools

In addition to main network devices, **utility tools** such as cables, crimping tools, etc are required.

Name	Price	Quantity	Link
Solsop Pass Through RJ45 Crimp Tool Kit All-in-One	\$ 35 (x2)	1 (x2)	Click Here
Cat 6 Ethernet Cable 200 ft	\$ 28 (x2)	1 (x2)	Click Here
RJ45 Connector Pack (100)	\$ 15 (x2)	5 (x2)	Click Here
Total	\$ 276		

Power Supply

Although the business will have a backup power supply, such as a generator, the network devices require an immediate UPS for uninterrupted operation. The estimated combined power usage of the router and two multilayer switches would be around **250W**, and the access switches would be **150W**.

APC 2200VA Smart UPS (1683 Wh) will be used for the router and multilayer switches, with estimated backup time of around 45-60 mins.

APC Smart-UPS 1500VA (1000 Wh) will supply the access switches for around 50-70 mins.

UPS model	Rated Capacity (Wh)	Backup Time	Quantity	Price	Link
APC Smart-UPS 1500VA	1000 Wh	50–70 min	1 (x2)	\$ 600 (x2)	Click Here
APC Smart-UPS 2200VA	1683 Wh	45–60 min	1 (x2)	\$ 1,175 (x2)	Click Here
Total			\$ 3,550		

Total Cost

Name	Type	Price
C1111-4P	Router	\$ 1,990
C9200L-24T-4G	Multilayer Switch	\$ 3,840
CBS250-48T-4X	L2 Switch	\$ 4,200
Utility Tools and Cables	miscellaneous	\$ 276
APC Smart-UPS	UPS	\$ 3,550
Miscellaneous Fees (such as shipping fees, installation fees)		\$ 500
Total (in USD)		\$ 14,356
Total (in MMK)		64,602,000 MMK

(Note: Exchange Rate as of *February 26th 2025* is *1 USD = 4,500 MMK*)

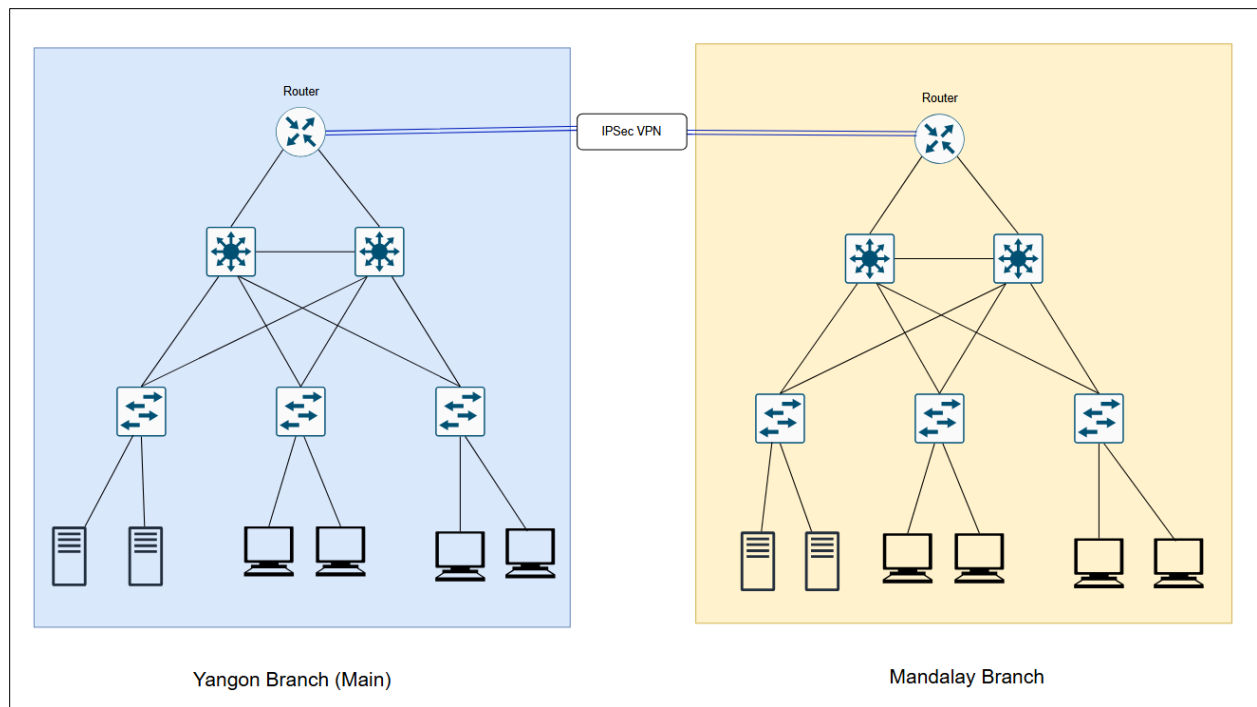


Fig 4.1.2 Entire Network Architecture of TipTop (with IPsec VPN)

4.2 Implementation of IPSec VPN

IPSec is a end-to-end security protocols suite that encrypts and authenticate data packets between two separate networks via the Internet, without any physical implementations. The framework includes set of protocols for mutual authentication, and negotiation of crypto keys to during sessions between the agents. By using it as *site-to-site VPN*, two business branches could communicate as if they were in the same location privately without exposing to the ISP or the Internet (Cisco, n.d.). IPSec can provide following features:

- Encrypt Data: use cryptographic means to encrypt data and protect it from being intercepted.
- Authenticate Communication: ensures only authenticated devices can communicate with each other
- Integrity: protects the integrity of the transmitted communications

4.2.1 IPSec Modes

It can operate in two modes, Tunnel Mode and Transport Mode.

Tunnel Mode: encrypts the entire data packet, which is ideal for site-to-site VPN, and

Transport Mode: encrypts only the data inside the packet

Two Main Protocols used in IPSec security protocols are:

- ESP (Encapsulating Security Payload): a protocol that transport encrypted and integrity verified data across the network. Authentication Header (AH) was the older method, used for IPSec transportation of unencrypted data, but it is no longer recommended.
- IKE (Internet Key Exchange): protocol used by IPSec to negotiate connection details, security parameters, and manage, update and delete information. As recommended in (Elaine Barker, 2020), Internet Key Exchange (IKEv2) is recommended. However, IKEv1 will be used for this project due to the limitations of the Packet Tracer. In addition to these protocols, IP payload compression protocol (IPComp) can be used for packet compression.

4.2.2 Additional Considerations

Although IPSec VPN could encrypt and provide some security, it could not eliminate all potential threats. One such example is flaw in encryption algorithm that could lead to attackers decrypting and intercepting the traffic. Moreover, in the attempt to cover all potential threats, the VPN tends to decrease availability due to the additional security features as they add more complexity and utilize more resources (Elaine Barker, 2020). As such, it is important to strike the balance between optimizing for availability and security considerations.

Site-to-site IPSec VPN will be used for communication between TipTop main branch and its side branch in different city. Algorithms used for encryption and integrity will be included in the Configuration section. (Tanenbaum, 2011) (Ross, 2022)

4.2.3 Configurations

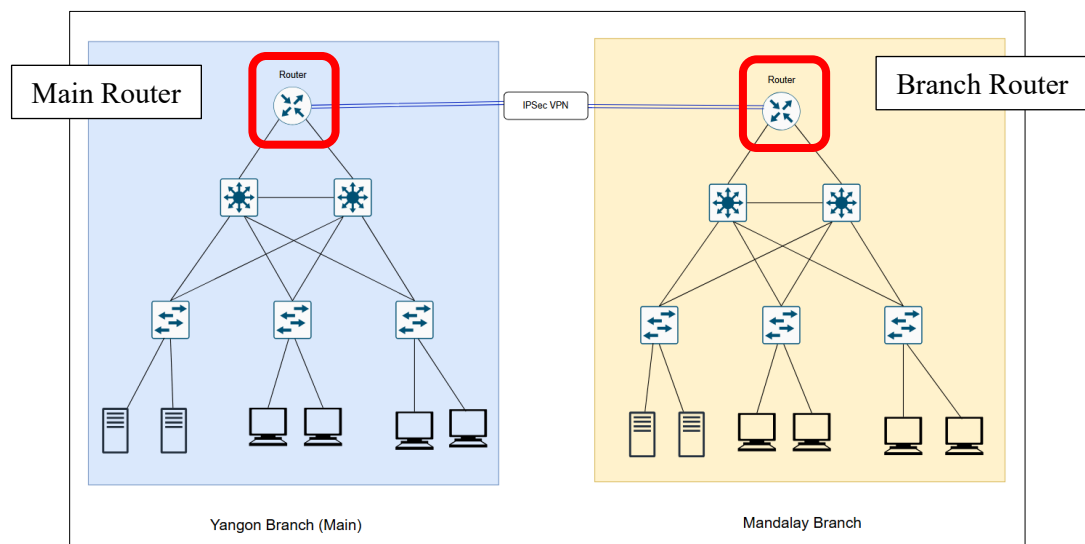


Fig 4.2.3.1 IPSec VPN Implementation

The IPSec VPN will be configured on the router of each branch as shown in Fig 4.2.3.1. The configuration is done in following order:

1. Access Control List
2. ISAKMP Policy (IKE Phase 1)
3. IPSec transform-set (Phase 2)
4. Crypto Map
5. Applying the crypto map to interface

Step 1: Access Control List

Configuration on Main Router

```
access-list 100 permit ip 10.0.0.0 0.0.3.255 20.0.0.0 0.0.3.255
access-list 100 permit ip 10.0.0.0 0.0.3.255 192.168.4.0 0.0.3.255
access-list 100 permit ip 192.168.0.0 0.0.3.255 20.0.0.0 0.0.3.255
access-list 100 permit ip 192.168.0.0 0.0.3.255 192.168.4.0 0.0.3.255
```

Configuration on Branch Router

```
access-list 100 permit ip 20.0.0.0 0.0.3.255 10.0.0.0 0.0.3.255
access-list 100 permit ip 20.0.0.0 0.0.3.255 192.168.0.0 0.0.3.255
access-list 100 permit ip 192.168.4.0 0.0.3.255 10.0.0.0 0.0.3.255
access-list 100 permit ip 192.168.4.0 0.0.3.255 192.168.0.0 0.0.3.255
```

Access List (ACL) defines which traffics should be encrypted by the IPSec VPN. **Extended ACL 100** is used on both routers. As shown in figure, on the *Main Router*, traffics from 10.0.0.0/22 and 192.168.0.0/22 networks to remote networks (TipTop Branch) are included in ACL and vice versa.

Step 2: ISAKMP Policy (IKE Phase 1)

```
Configuration on Main Router
crypto isakmp policy 10
  encr aes 256
  authentication pre-share
  group 5
!
crypto isakmp key secretkey address 20.0.0.2

Configuration on Branch Router
crypto isakmp policy 10
  encr aes 256
  authentication pre-share
  group 5
!
crypto isakmp key secretkey address 10.0.0.2
```

ISAKMP (Internet Security Association and Key Management Protocol) is essential for security negotiations in establishing IPsec VPN.

ISAKMP policy 10 is firstly created. AES-256 encryption is selected for its strong security. Pre-shared Key method (PSK) will be used for authentication. **Diffie-Hellman Group 5** is used for key exchange.

On *Main Router*, ‘secretkey’ will be used as the pre-share key for authentication with the VPN peer at 20.0.0.2, and vice versa. The policy facilitates how the VPN peers will communicate and exchange the encryption keys securely. The pre-shared key must match on both routers for the VPN tunnel to be established.

Step 3: IPsec transform-set (Phase 2)

```
Configuration on Main Router
crypto ipsec transform-set R1-R3 esp-aes 256 esp-sha-hmac

Configuration on Branch Router
crypto ipsec transform-set R3-R1 esp-aes 256 esp-sha-hmac
```

As shown in figure, transform-set ‘R1-R3’ is configured on the *Main Router*, and ‘R3-R1’ on *Branch Router*. This defines how the data will be encrypted and authenticated once the VPN is established. **AES-256** algorithm is used for encryption, and **SHA-HMAC** is used for integrity and authentication. (Note: aside from the name, all the settings must match)

Step 4: Crypto Map

```
Configuration on Main Router
crypto map IPSEC-MAP 10 ipsec-isakmp
 set peer 20.0.0.2
 set pfs group5
 set security-association lifetime seconds 86400
 set transform-set R1-R3
 match address 100

Configuration on Branch Router
crypto map IPSEC-MAP 10 ipsec-isakmp
 set peer 10.0.0.2
 set pfs group5
 set security-association lifetime seconds 86400
 set transform-set R3-R1
 match address 100
```

In this step, crypto map 10 is created and mapped to ISAKMP. *Branch Router (20.0.0.2)* is set as VPN peer on *Main Router*, and vice versa. After that, Perfect Forward Secrecy (PFS) with Diffie-Hellman Group 5 is enabled. **VPN SA (Security Association)** is set to expire after 24 hrs (86400s). 'R1-R3' which was defined in previous step is set as the transformed set on *Main Router*, and 'R3-R1' is set as transformed set on the *Branch Router*. Traffic matching the ACL 100 will be encrypted and sent through the IPsec VPN.

Step 5: Applying Crypto map to Interface

```
int g0/0
crypto map IPSEC-MAP
```

For the VPN tunnel to be created, it must be applied to an interface connecting to the peer (for example, g0/0). This enables IPsec VPN encryption on the interface g0/0, and all outgoing traffic matching ACL 100 will be sent over the VPN.

Verification Commands and Testing

Following commands can be used to verify if the IPSec tunnel is working as intended.

```
- sh crypto ipsec sa (shows active VPN tunnels and statistics)
- sh crypto isakmp policy (Lists configured ISAKMP policies)
- sh crypto ipsec transform-set (Displays encryption and authentication settings)
```

Fig 4.2.3.2 common verification commands for IPSec VPN

For example, PC in admin department 192.168.2.6/24 wishes to communicate with a PC in branch office 192.168.4.6/24, following packet with actual source and destination IP addresses will be observable as shown in figure below.

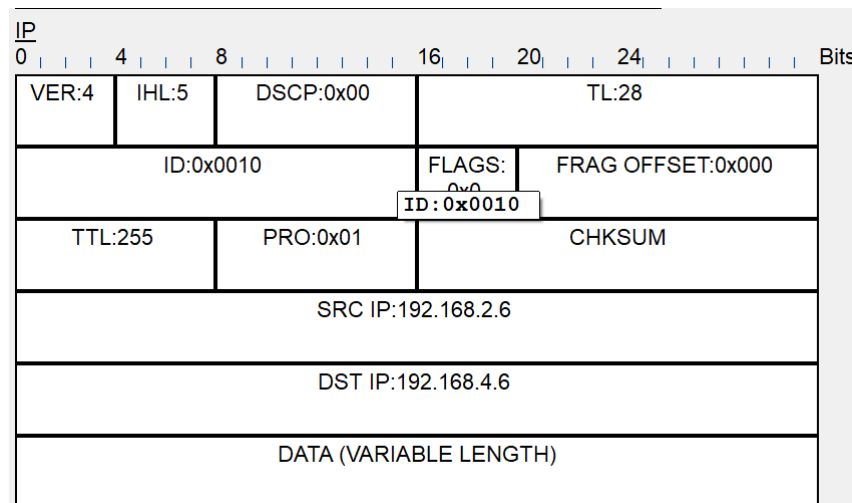


Fig 4.2.3.3 Unencrypted packet inside the Main Branch network

After the packet reaches the gateway router where the IPSec VPN is configured, it will encrypt the packet and send to the peer router on the branch site. During this transmission between two peer routers, to the outside network (such as the ISP and other devices), following masked packet header will be shown.

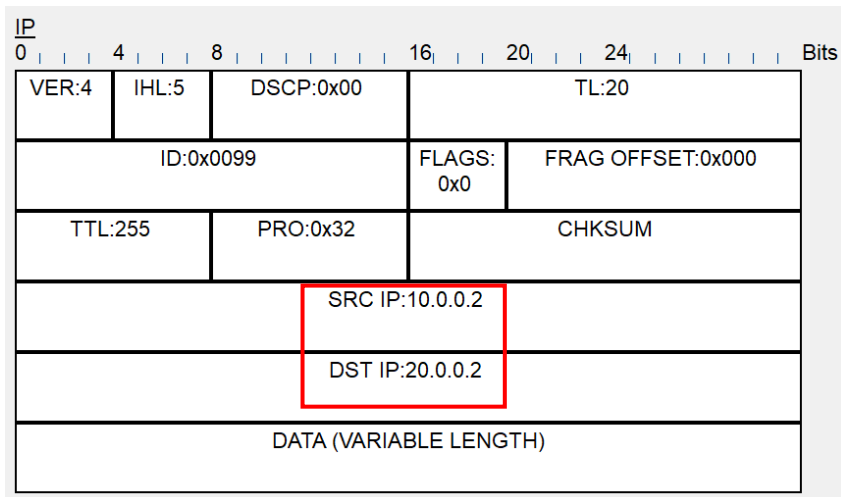


Fig 4.2.3.4 Encrypted packet to the outside network

Outside network devices (such as the ISP) are led to think the packets are intended for each router on each end as the source and destination IP addressees of the packets, and only with the key will it be able to unencrypt the packets.

Once the packet reached the peer router on the branch network, the router will decrypt the packet and inspect the integrity of the packet. It will then route the packet to the internal branch network to the actual destination PC 192.168.4.6/24.

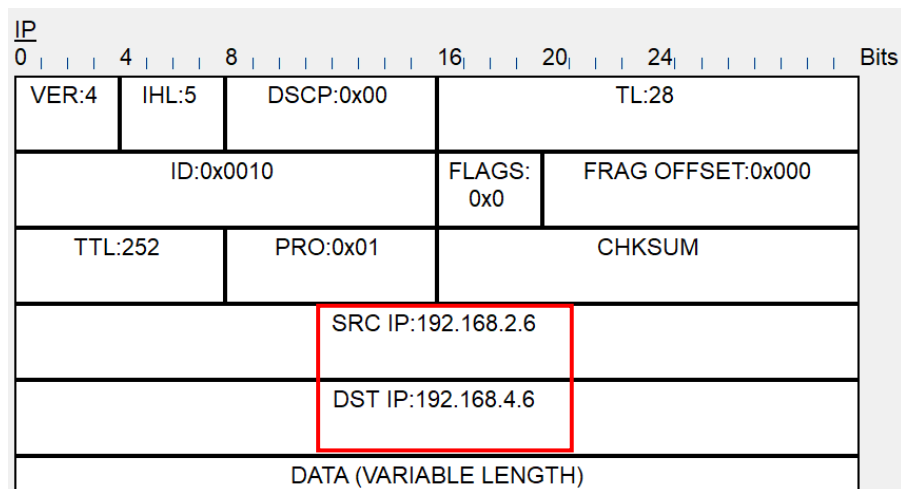


Fig 4.2.3.5 Decrypted and Integrity ensured packet in the Branch network

During this example transmission of a packet, the outside network is unaware of the actual contents nor the source and destination of the packet. Once the packet has reached to other end of the VPN tunnel, it will check the integrity of the packet to ensure contents of the packet remain intact, thus preventing from potential interceptions and alterations.

Checking SA on Main Router

```
R1(config)#do sh crypto ipsec sa

interface: GigabitEthernet0/0
  Crypto map tag: IPSEC-MAP, local addr 10.0.0.2

  protected vrf: (none)
  local  ident (addr/mask/prot/port): (10.0.0.0/255.255.252.0/0/0)
  remote ident (addr/mask/prot/port): (20.0.0.0/255.255.252.0/0/0)
  current_peer 20.0.0.2 port 500
    PERMIT, flags={origin_is_acl,}
    #pkts encaps: 0, #pkts encrypt: 0, #pkts digest: 0
    #pkts decaps: 0, #pkts decrypt: 0, #pkts verify: 0
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0

    local crypto endpt.: 10.0.0.2, remote crypto endpt.:20.0.0.2
    path mtu 1500, ip mtu 1500, ip mtu idb GigabitEthernet0/0
    current outbound spi: 0x0(0)

  inbound esp sas:

--More--
```

Checking SA on Mandalay Branch Router

```
R2#sh crypto ipsec sa

interface: GigabitEthernet0/0
  Crypto map tag: IPSEC-MAP, local addr 20.0.0.2

  protected vrf: (none)
  local  ident (addr/mask/prot/port): (20.0.0.0/255.255.252.0/0/0)
  remote ident (addr/mask/prot/port): (10.0.0.0/255.255.252.0/0/0)
  current_peer 10.0.0.2 port 500
    PERMIT, flags={origin_is_acl,}
    #pkts encaps: 0, #pkts encrypt: 0, #pkts digest: 0
    #pkts decaps: 0, #pkts decrypt: 0, #pkts verify: 0
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0

    local crypto endpt.: 20.0.0.2, remote crypto endpt.:10.0.0.2
    path mtu 1500, ip mtu 1500, ip mtu idb GigabitEthernet0/0
    current outbound spi: 0x0(0)

  inbound esp sas:

--More-- |
```

Checking IKE Policy on Main Router

```
R1(config)#do sh crypto isakmp policy

Global IKE policy
Protection suite of priority 10
  encryption algorithm:      AES - Advanced Encryption Standard (256 bit keys).
  hash algorithm:            Secure Hash Standard
  authentication method:     Pre-Shared Key
  Diffie-Hellman group:      #5 (1536 bit)
  lifetime:                  86400 seconds, no volume limit
```

Checking IKE Policy on Branch Router

```
R2#sh crypto isakmp policy

Global IKE policy
Protection suite of priority 10
  encryption algorithm:      AES - Advanced Encryption Standard (256 bit keys).
  hash algorithm:            Secure Hash Standard
  authentication method:     Pre-Shared Key
  Diffie-Hellman group:      #5 (1536 bit)
  lifetime:                  86400 seconds, no volume limit
R2#
```

4.3 Implementation of OSPF

OSPF (Open Shortest Path First) is a linked-state dynamic routing protocol with Dijkstra's Shortest Path algorithm to calculate the shortest and most efficient path between network devices.

Key characteristics of OSPF include:

- OSPF has a hierarchical network design using *Areas*
- *Neighbour relationship* will be established between OSPF configured routers in the same Area
- If there were any changes, OSPF uses **LSA (Link-state Advertisements)** to send only the changes
- LSAs are refreshed between neighbours every 30 minutes
- OSPF uses multicast addresses of
 - o 224.0.0.5 (all OSPF configured routing devices)
 - o 224.0.0.6 (all designated routers)
- OSPF has Administrative Distance (AD) of 110 and uses *cost* as its metric, which is calculated based on the bandwidth of the links
- Highly scalable as it has no hop-count limit (Balchunas, 2007)

4.3.1 OSPF Neighbours

Each OSPF router is determined by Router ID, and forms neighbour relationship called adjacencies with routers in the same *Area* by sending Hello message, which contains Area ID, Area Type, Prefix, Hello and Dead interval and other parameters.

After the adjacencies are established, a neighbour table is created using information from Hello messages. The table contains information about Router ID, current State of the neighbour, interfaces connected, and the IP addresses of connected interface of the neighbours (Balchunas, 2007).

4.3.2 Designated OSPF router

Each OSPF devices will send LSA of their own to establish adjacencies, and causes LSA flooding which could lead to network congestions. To resolve this issue, OSPF will elect a **Designated Router (DR)**, communicated via 224.0.0.6, and a **Backup Designated Router (BDR)** for redundancy in case of DR failure. The rest of the non-DR/BDR OSPF devices are called DR-Others. In addition, the DR and BDR election is non pre-emptive, meaning there will be no changes until the network resets.

The election of DR and BDR is determined by the OSPF Priority of the router. The Priority of a router can be manually configured on the interface, and if priorities are a tie, it will elect based on following order:

Router ID → Loopback Address → IP address of highest UP interface

The DR as the primary distributor of LSAs, it will send LSA to every other DR-Others to ensure they have the identical **link-state databases (LSDB)**, which stores the LSAs (networkacademy, n.d.).

4.3.3 OSPF Neighbour States

OSPF configured devices will go through following process to establish adjacencies:

Down: does not aware of any neighbouring OSPF devices (no Hello messages)

Init: receives Hello message but own router ID is not in the packet

2 Way: bidirectional communication has been established. The Hello packet contains both own's RouterID and neighbours RouterID.

Exstart: routers are preparing to share information about their LSDB; router with higher Router ID will become **Master**, and will initiate the exchange. Lower Router ID will become the **Slave**.

Exchange: routers exchange DBDs, which contains list of LSA in their LSDB. After the exchange, the router will compare the LSDBs to synchronize the LSAs between the neighbours.

Loading: in this state, the routers send LSR (link-state request) to request LSAs they don't have from their neighbours.

Full: all the neighbours in the OSPF Area have full adjacency and identical LSDBs. Hello packet is sent every 10 second to maintain the adjacency. If no Hello packet is received within 40 second (Dead Timer), the neighbour is removed. (Balchunas, 2007) (Jeremy's IT Lab, 2021)

4.3.4 OSPF Network Types

Broadcast Multi-Access: default network type for Ethernet, Token Ring and ATM. OSPF will elect DR and BDR.

Point-to-Point: usually used in point-to-point connections. No DR/BDR election is needed, and all traffic multicasts to 224.0.0.5.

Non-broadcast Multi-access Network: one interface can connect to multiple destinations, but broadcast cannot be sent on NBMA network. DR and BDR will be elected but all the neighbours must be manually defined (Balchunas, 2007).

There are OSPF network types but will not be mentioned in this documentation.

4.3.4 Configurations

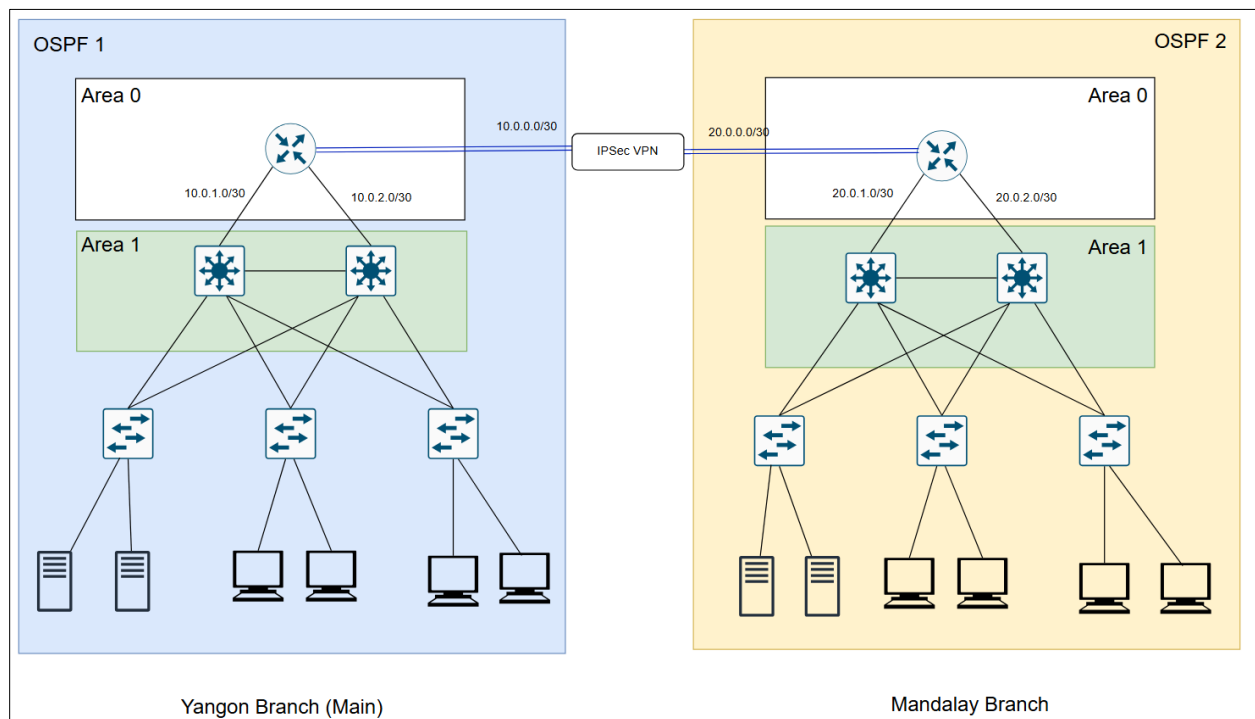


Fig 4.3.4.1 OSPF configured L3 devices

In this project, Broadcast Multi-Access OSPF type will be used on the layer-3 devices as shown in figure 4.3.4.1. This includes the two main routers (one on each branch), and the multilayer switches in the Distribution Layer of each branch. However, between the two IPsec routers, no GRE tunnel is used.

Configuration on Main Branch Router

```
router ospf 1
 log-adjacency-changes
 network 10.0.1.0 0.0.0.3 area 0
 network 10.0.2.0 0.0.0.3 area 0
 default-information originate
```

On the Main branch router, OSPF Process ID 1 is enabled, and networks connecting to the multilayer switches (10.0.1.0/30 and 10.0.2.0/30) are advertised to Area 0. As the gateway of the Main Branch LAN, it injects the default route (0.0.0.0/0) to the OSPF domain for communication with external network.

Configuration on Mandalay Branch Router

```
router ospf 2
 log-adjacency-changes
 network 20.0.1.0 0.0.0.3 area 0
 network 20.0.2.0 0.0.0.3 area 0
 default-information originate
```

Similarly, on the Branch Router, OSPF Process 2 is enabled, and 20.0.1.0/30 and 20.0.2.0/30 networks are advertised in Area 0. The Branch Router, also being the gateway to Branch LAN, injects the default route (0.0.0.0/0) to OSPF domain for external communication.

Configuration on Main Branch Multilayer Switches

```
router ospf 1
 log-adjacency-changes
 network 10.0.1.0 0.0.0.3 area 0
 network 192.168.1.0 0.0.0.255 area 1
 network 192.168.2.0 0.0.0.255 area 1
 network 192.168.3.0 0.0.0.255 area 1
```

On the two Multilayer switches in the Main Branch LAN, OSPF Process 1 is enabled, and Admin, Servers, Sales and Server VLANs (such as 192.168.1.0/24, 192.168.2.0/24) are advertised in Area 1. The network connecting to the gateway router (10.0.0.1/30 and 10.0.2.0/30) are advertised in area 0. Additional networks can be advertised in case of business expansion or increase in the number of users in the future.

Configuration on Main Multilayer Switches

```
router ospf 2
 log-adjacency-changes
 network 20.0.1.0 0.0.0.3 area 0
 network 192.168.4.0 0.0.0.255 area 1
 network 192.168.5.0 0.0.0.255 area 1
 network 192.168.6.0 0.0.0.255 area 1
```

OSPF with Process ID 2 is enabled in the Mandalay Branch multilayer switches. Similar to their counterparts in Main Branch, department networks (such as 192.168.4.0/24, 192.168.5.0/24) are advertised in Area 1, while the network with the router (20.0.1.0/30 and 20.0.2.0/30) are advertised in Area 0.

Verification Commands

Following are OSPF verification commands.

```
- sh ip ospf neighbor           (list OSPF neighbours, State and Interface)
- sh ip ospf neighbor detail    (neighbour details DR/DBR, uptime, deadtimer)
- sh ip ospf interface          (checking OSPF interfaces)
- sh ip ospf int br             (summarizes OSPF interfaces and their status)

- sh ip route ospf              (checking OSPF routes)
- sh ip ospf database            (checking LSDBs)
```

Checking OSPF routes on Main Branch Router

```
R1(config)#do sh ip ro ospf
O IA 192.168.1.0 [110/2] via 10.0.2.2, 00:00:36, GigabitEthernet0/2
[110/2] via 10.0.1.2, 00:00:36, GigabitEthernet0/1
O IA 192.168.2.0 [110/2] via 10.0.2.2, 00:00:36, GigabitEthernet0/2
[110/2] via 10.0.1.2, 00:00:36, GigabitEthernet0/1
O IA 192.168.3.0 [110/2] via 10.0.2.2, 00:00:36, GigabitEthernet0/2
[110/2] via 10.0.1.2, 00:00:36, GigabitEthernet0/1
```

Checking Main Branch Router LSDB

```
R1(config)#do sh ip ospf database
      OSPF Router with ID (10.0.2.1) (Process ID 1)

      Router Link States (Area 0)

Link ID        ADV Router    Age         Seq#          Checksum Link count
192.168.3.2    192.168.3.2   105        0x80000003   0x00d189 1
10.0.2.1       10.0.2.1      105        0x80000006   0x007e71 2
192.168.3.3    192.168.3.3   105        0x80000003   0x00d77f 1

      Net Link States (Area 0)

Link ID        ADV Router    Age         Seq#          Checksum
10.0.1.2       192.168.3.2   105        0x80000001   0x0025ca
10.0.2.2       192.168.3.3   105        0x80000001   0x0013da

      Summary Net Link States (Area 0)

Link ID        ADV Router    Age         Seq#          Checksum
192.168.1.0    192.168.3.3   100        0x80000001   0x00a3e0
192.168.1.0    192.168.3.2   100        0x80000001   0x00a9db
192.168.2.0    192.168.3.3   100        0x80000002   0x0096eb
192.168.2.0    192.168.3.2   100        0x80000002   0x009ce6
192.168.3.0    192.168.3.3   100        0x80000003   0x0089f6
192.168.3.0    192.168.3.2   100        0x80000003   0x008ff1

      Summary ASB Link States (Area 0)

Link ID        ADV Router    Age         Seq#          Checksum
10.0.2.1       192.168.3.2   80         0x80000004   0x00af2f
10.0.2.1       192.168.3.3   80         0x80000004   0x00a934
--More--
```

Checking LSDB on Mandalay Branch Multilayer Switch

```
Switch#sh ip ospf data
      OSPF Router with ID (192.168.6.2) (Process ID 2)

      Router Link States (Area 0)

Link ID      ADV Router    Age      Seq#          Checksum Link count
192.168.6.3  192.168.6.3    87       0x80000003   0x0056e6 1
192.168.6.2  192.168.6.2    87       0x80000003   0x0050f0 1
20.0.2.1     20.0.2.1       87       0x80000006   0x00fbb7 2

      Net Link States (Area 0)

Link ID      ADV Router    Age      Seq#          Checksum
20.0.2.2     192.168.6.3    87       0x80000001   0x00f8eb
20.0.1.2     192.168.6.2    87       0x80000001   0x003ba7

      Summary Net Link States (Area 0)

Link ID      ADV Router    Age      Seq#          Checksum
192.168.4.0  192.168.6.3    82       0x80000001   0x006d11
192.168.5.0  192.168.6.3    82       0x80000002   0x00601c
192.168.6.0  192.168.6.3    82       0x80000003   0x005327
192.168.4.0  192.168.6.2    77       0x80000001   0x00730c
192.168.5.0  192.168.6.2    77       0x80000002   0x006617
192.168.6.0  192.168.6.2    77       0x80000003   0x005922

      Summary ASB Link States (Area 0)

Link ID      ADV Router    Age      Seq#          Checksum
20.0.2.1     192.168.6.3    77       0x80000004   0x0012be
20.0.2.1     192.168.6.2    72       0x80000004   0x0018b9

      Router Link States (Area 1)

Link ID      ADV Router    Age      Seq#          Checksum Link count
192.168.6.2  192.168.6.2    87       0x80000007   0x00bfe5 3
192.168.6.3  192.168.6.3    87       0x80000007   0x00fda2 3
```

Checking OSPF routes on Main Branch Multilayer Switch

```
Switch#sh ip ro ospf
      10.0.0.0/30 is subnetted, 2 subnets
O       10.0.1.0 [110/2] via 10.0.2.1, 00:11:06, FastEthernet0/1
O*E2    0.0.0.0/0 [110/1] via 10.0.2.1, 00:11:06, FastEthernet0/1
```

Checking OSPF routes on Mandalay Branch Multilayer Switch

```
Switch#sh ip ro ospf
      20.0.0.0/30 is subnetted, 2 subnets
O       20.0.2.0 [110/2] via 20.0.1.1, 00:03:37, FastEthernet0/1
O*E2    0.0.0.0/0 [110/1] via 20.0.1.1, 00:03:37, FastEthernet0/1
```

Tracing route from Main Branch PC to Mandalay Branch PC (192.168.4.7)

```
C:\>tracert 192.168.4.7

Tracing route to 192.168.4.7 over a maximum of 30 hops:

  1  0 ms      0 ms      0 ms      192.168.3.2
  2  0 ms      0 ms      0 ms      10.0.2.1
  3  *         0 ms      0 ms      20.0.0.2
  4  0 ms      0 ms      0 ms      20.0.2.2
  5  0 ms      0 ms      0 ms      192.168.4.7
```

4.4 Redundancy and HSRP

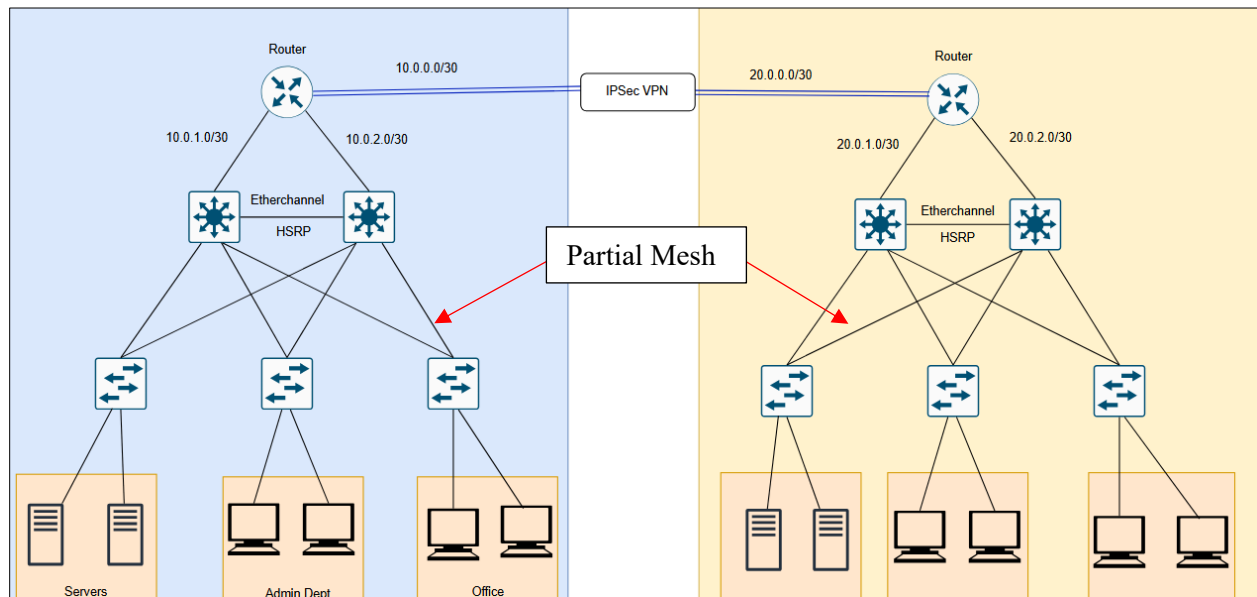


Fig 4.4.1 Redundancy in Distribution Switches, and Partial Mesh in Access Switches

Redundancy is critical for ensuring high availability and reduce downtime; one important aspect of redundancy is to maintain seamless connectivity even when primary device fails. For TipTop network, HSRP (Hot Standby Router protocol) and EtherChannel (LACP) will be implemented in distribution layer of both branches. Additionally, access layer switches are connected to the distribution switches in partial-mesh topology, further enhancing redundancy and network resilience.

4.4.1 EtherChannel (LACP)

EtherChannel aggregates many physical ports into one logical link, providing higher bandwidth and redundancy. There are two EtherChannel protocols; Cisco's proprietary Port Aggregation Protocol (PAgP), and IEEE standard Link Aggregation Control Protocol (LACP). For this project, LACP will be used.

Configuring EtherChannel (LACP) on distribution switches of both Branches

```
int range f0/5-6
channel-protocol lacp          (LACP) link aggregation control protocol
channel-group 1 mode active    (activating LACP unconditionally)

sh etherchannel summary        (checking etherchannel status)
```

Checking EtherChannel Status using Verifying command

```
Switch(config-if-range)#do sh ether sum
Flags:  D - down          P - in port-channel
        I - stand-alone  s - suspended
        H - Hot-standby (LACP only)
        R - Layer3        S - Layer2
        U - in use        f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

Number of channel-groups in use: 1
Number of aggregators:          1

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Po1 (SU)        LACP       Fa0/5 (P) Fa0/6 (P)
```

4.4.2 HSRP (Hot Standby Router protocol)

Cisco-proprietary First Hop Redundancy Protocol (FHRP) that allows two or more layer-3 devices to work together as a single virtual router, providing a backup gateway. One router in HSRP will become *the active router* and the other one, *standby router* based on priority; both routers sharing one Virtual IP Address. HSRPv1 will be implemented on the distribution layer switches of both Branches.

Following HSRP configurations are applied to L3 switches of both Yangon and Mandalay Branches. Configurations on Yangon Branch L3 switches are shown below.

Configuring HSRP on Active L3 switch of Yangon Branch

```
int vlan 10                                (selecting interface)
standby 1 ip 192.168.1.1                   (setting hsrp virtual ip address)
standby 1 priority 110                     (setting higher priority on Active router)
standby 1 preempt                           (enabling preemption)
```

Configuring HSRP on Standby L3 switch of Yangon Branch

```
int vlan 10                                (selecting interface)
standby 1 ip 192.168.1.1                   (setting hsrp virtual ip address)
standby 1 priority 110                     (setting lower priority on Standby router)
```

Verifying HSRP

```
Switch#sh standby br
P indicates configured to preempt.
|
Interface    Grp  Pri P State    Active        Standby        Virtual IP
Vl10         10   110 P Active   local         192.168.1.3    192.168.1.1
Vl20         20   110 P Active   local         192.168.2.3    192.168.2.1
Vl30         30   110 P Active   local         192.168.3.3    192.168.3.1
```

4.5 Remote Access

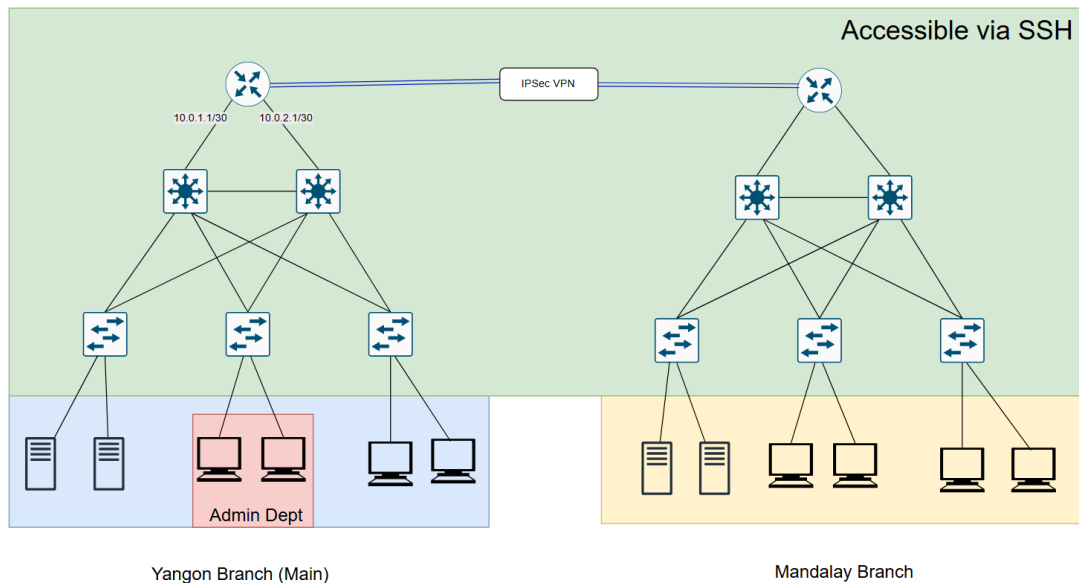


Fig 4.5.1 Network devices of both Branches (green) accessible via SSH from Admin Dept (red)

Remote Access allows administrators to access and manage network devices from a different location without any physical access. Nowadays, Secure Shell (SSH) is widely for remote access, replacing methods such as Telnet. In contrast to Telnet which sends data in plaintexts, SSH encrypts communication, and uses cryptographic authentication for secure logins.

Implementation of Remote Access capability in TipTop network would allow admins to manage devices both Yangon and Mandalay branches from a single location, **eliminating the need for travel between sites.**

Configuration of SSH

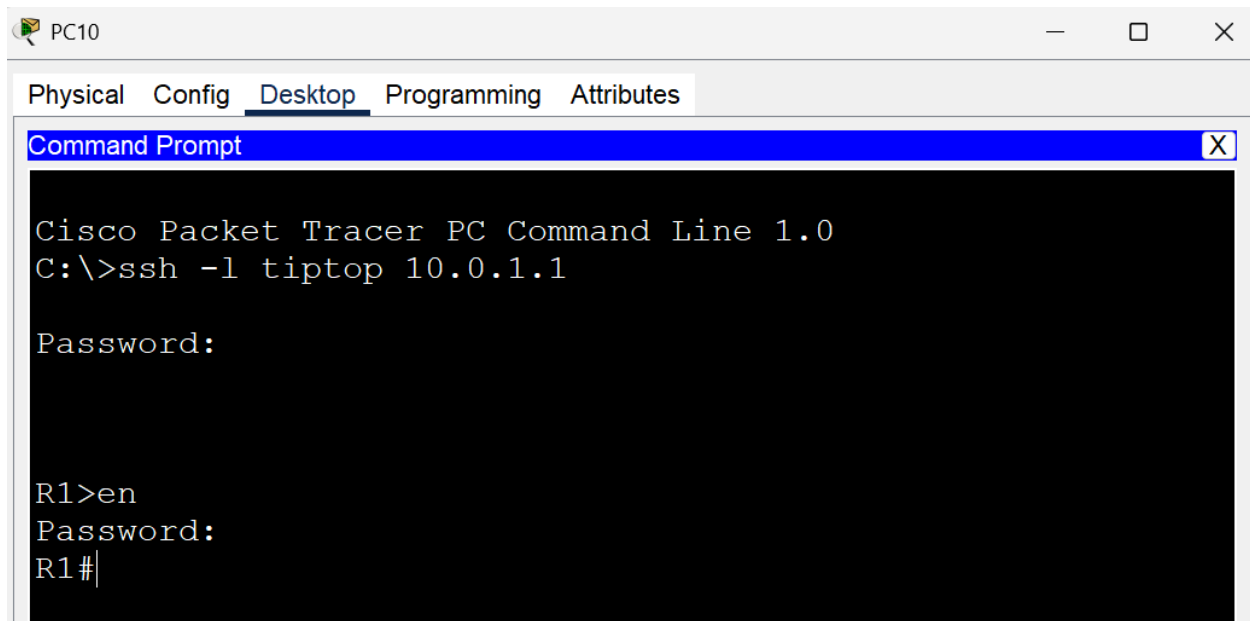
```
-username tiptop password 12345          (setting connection username and password)

-hostname R1
-ip domain name mydomain.com
-crypto key generate rsa general-key modulus 1024      (generate cryptographic key
-enable password 000000                                (password to go into en mode
-line vty 0 4                                           (allowing 5 simultaneous ssh connections
-login local
-transport input ssh
```

SSH login commands for Admin PC

```
ssh -l tiptop 10.0.2.1          (ssh login, username, ip address)
password: 12345                 (entering password)
```

Accessing the Main Branch Router from Admin Dept via SSH



4.6 VLAN and DHCP

VLAN

Virtual LAN (VLAN) is a logical subnet of network that divides a group of devices into separate broadcast domains, improving security, broadcast traffic, and simplifying management. The IP Addresses and names of the VLAN will be assigned as below using Dynamic Host Configuration Protocol (DHCP).

	VLAN	IP Address	Name	Number of Devices
Yangon Branch	VLAN-10	192.168.1.0/24	Servers	255 devices
	VLAN-20	192.168.2.0/24	Admin Department	
	VLAN-30	192.168.3.0/24	Office Users	
Mandalay Branch	VLAN-40	192.168.4.0/24	Vlan40	
	VLAN-50	192.168.5.0/24	Vlan50	
	VLAN-60	192.168.6.0/24	Vlan60	

DHCP

DHCP automatically assign IP addresses and other network configuration (such as subnet mask, default gateway, DNS servers) to devices on a network, eliminating the need for manual IP address assignment.

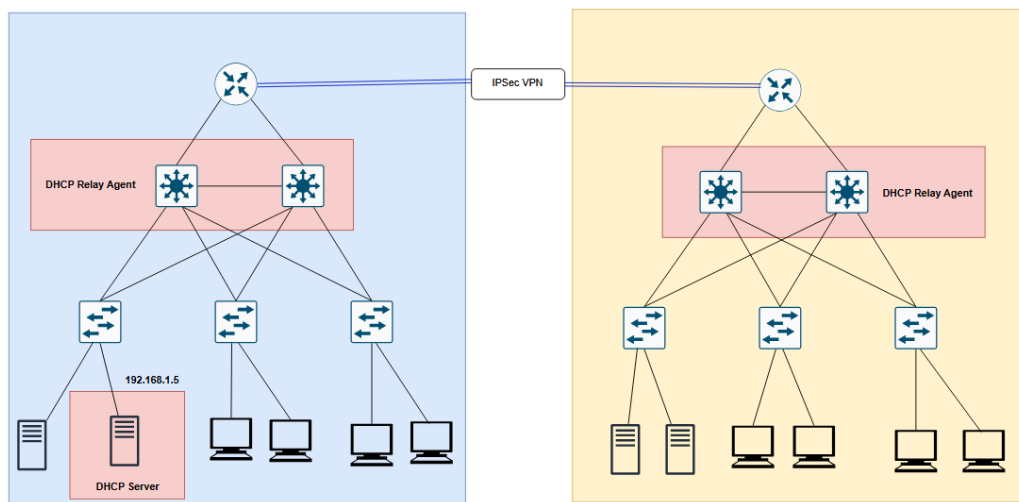


Fig 4.6.1 DHCP Server and relay agents

Configuration of DHCP Server

```
ip dhcp pool <name> (setting DHCP pool name)
network 192.168.1.0 255.255.255.0 (DHCP network)
default-router 192.168.1.1 (setting default gateway of the DHCP network)
dns-server 8.8.8.8 (IP address of DNS server, might be same as DHCP server)
```

Configuration of DHCP relay agents

```
int vlan10 (interface connecting to the VLAN)
ip helper-address <ip of the dhcp server>
```

Verifying DHCP on Main Branch PC

IP Configuration	
☒ DHCP	☐ Static
IPv4 Address	192.168.2.6
Subnet Mask	255.255.255.0
Default Gateway	192.168.2.1
DNS Server	192.168.1.5

5. Conclusion

5.1 Limitation

Firstly, although other VPN protocols such as Wireguard are more efficient and provide stronger encryptions, IPSec VPN has been chosen due limitations of the Packet Tracer and other technical issues. In addition, technologies such as Port Security, Asymmetric ACL and IKEv2 are not supported in the Packet Tracer.

Secondly, VoIP and QoS were initially planned for implementation; however, it was later discovered that routers used in this project do not support telephony and QoS services. As such, the plan was dropped due to time constraints.

Lastly, using other advanced simulation tools such as GNS3 or EVE-ng should be considered for future implementation as Packet Tracer, although light-weight and user friendly, is severely limited in *advanced* features.

5.2 Future Considerations

While current implementation focuses on stable and secure network infrastructure with IPSec VPN connectivity between two branches to TipTop several technologies and advanced features were ultimately deferred due to time and resource constraints.

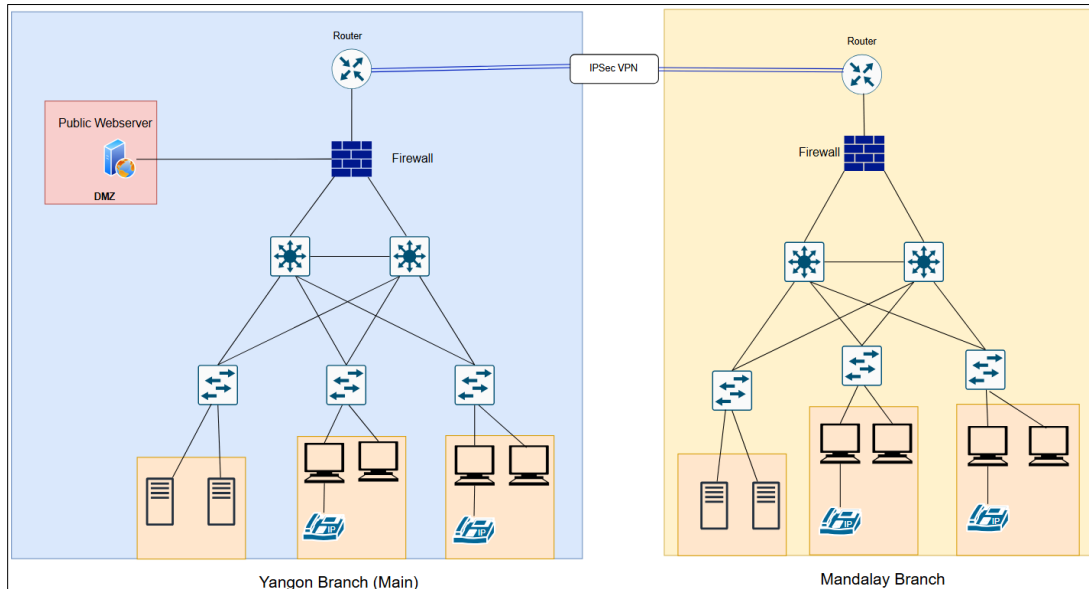


Fig 5.2.1 Future Network Architecture with Firewall, DMZ and additional features

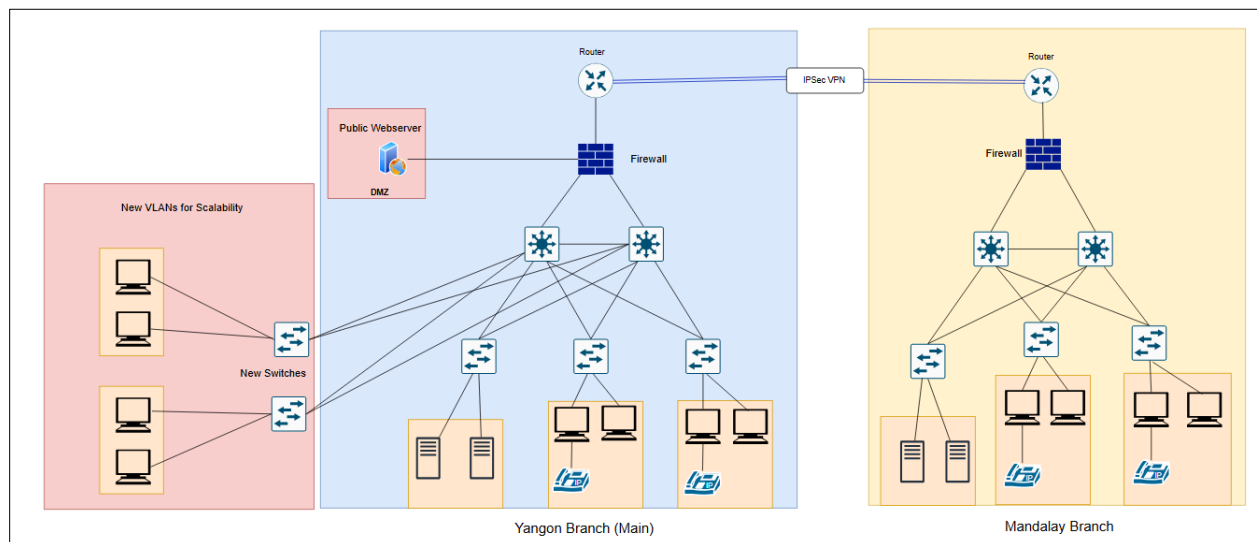


Fig 5.2.2 TipTop Network Architecture with Scalability for future business expansions

One primary consideration for future implementation is the addition of firewalls and Demilitarised Zone (DMZ). The firewall would provide granular control over network traffics, while the DMZ would allow the business to host publicly accessible web-servers without exposing the internal network.

Another initially considered feature was the VoIP and Quality of Service (QoS) which were scrapped due to time constraints, and simulator limitations. Integration of these technologies to existing network architecture could streamline communication between two branches, and QoS mechanisms would prioritize voice traffics, improving the call quality.

5.3 Conclusion

Implementation of 2-tier network infrastructure with IPSec VPN marks a significant step up for establishing a secure and reliable communication environment for TipTop. This solution efficiently and cost-effectively provides a secure communication channel between Yangon and Mandalay TipTop branches. Moreover, use of enterprise-grade devices, redundancy features and backup power supplies allow improved manageability and streamline the workflow even during disruptions and blackouts.

Ultimately, although advanced features such as firewalls, DMZ and VoIP are deferred for the future, this implementation of TipTop infrastructure not only addresses current operational issues, it also sets a solid foundation for future expansion and digitization of the business.

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